Comments on Shanahan

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1. Can an LLM feel pain?
2. What exactly are you asking?
3. I can say: When I have pain, I have this characteristic feeling, this sensation, which is individuated by how it feels, by its phenomenology.
    a. And I want to know whether an LLM has *that.*
4. People have tried any number of phrases to focus attention on this phenomenal property:
    a. there is something that *it's like to be in pain*; or sensations are unique in that one cannot be in them without being aware that one is in them; and so on.
5. Descartes started the trend of saying that, if you think about it, these sensations are *'private'* in a very strong sense:
        i. To you your own sensations are *self-intimating* in a way that they are not to others.
        ii. You are *infallible* about them in a way that others couldn't be.
        iii. You have a *directness* of access to them that others could not have.
    b. This grounds a thesis of strong, modalized, asymmetry between your access to your own sensations and that of others.
    c. These features have been disputed.
    d. One thing is clear: these sensations are real. We can debate whether they have these strong privacy features.
    e. Another thing is clear. Even if they don't have these strong modal privacy features, at the moment, they are at least private in a contingent sense: perhaps in principle someone can gain as good access to my sensations as I have, but at the moment I have strongly asymmetric privileged access.
6. Given this perhaps merely contingent privacy, it's very natural to think that our sensation terms refer to these phenomenologically individuated events: 'pain' refers to the sensation of being in pain.
    a. Given that fact, our question about LLMs seems like a perfectly factual question: do they have these feelings, do they have states with this kind of phenomenology?
    b. Not only does it seem factual; it doesn't even seem vague in the way most questions can be: there is either the sensation or there isn't.
    c. It may be very difficult to answer this question in a definitive way; it may even be impossible to do so.
    d. But the question seems like a perfectly factual one.
7. Murray wants to say, following Witt, that our question is not a factual question.
    a. There is no fact of the matter about whether LLMs feel pain.
    b. It calls for a *decision* on our part whether to extend sensation talk to these entities, something that will reflect an 'existential orientation.'
8. This thesis could be meant in one of two ways:
    a. It could be a view just about mental terms, and some other specific categories of concept. The view would be that mental terms have 'open texture.'
        i. That is, their meaning does not determine a fully determinate yes/no verdict as to whether something is or is not in their extension.
    b. Or it could be part of a very general view of language according to which all terms are like that.
    c. I couldn't always tell which of these views Murray had.
    d. Let's suppose it's the weaker and more defensible view.
9. Why does he hold that pain talk has an 'open texture' when it comes to AI systems?
    a. Doesn't the story I told about how I attach 'pain' to a specific type of sensation fix a determinate extension?

10. The Wittgensteinian answer is, no: We know from the 'private language argument' that that story about how sensation terms get their meaning is false.
    a. I have never seen a version of the PLA that I thought was any good, so I am not moved by that. (That includes Kripke's sophisticated version.)
    b. A version that M seems to like goes like this.
        i. Suppose you have a sensation type that you haven't had before; you focus your attention on the distinctive way in which it feels, and you call sensations of that type S.
        ii. A week later you have what seems to you to be the same sensation type occurring again, so you think: Ah, S again.
        iii. Witt says: What could be your criterion for whether it was S again or not? Your sensation is private, so no one else can say. You would be the only authority, so whatever you say would have to be deemed correct.
        iv. However, if whatever you say goes, then that only means that we can't here talk about correct and incorrect.
        v. So that means no determinate extension was fixed via this private procedure.
        vi. And that's why we need to look elsewhere for how the meaning of pain talk is fixed.
        vii. And here we get a quasi-behaviorist story, acc to which although pain talk isn't a name for some kind of behavior, still it is only at home in a setting in which there are embodied creatures who behave roughly as we do.
        viii. So that's why, if we are to extend our talk to disembodied creatures or differently bodied creatures, a decision is called for.
11. I don't really understand the alternative picture of pain language. But I also don't think the case has been made that we need it. Nothing has been shown to be wrong with the private procedure. The flaw in this argument is the transition from iii to iv.
    a. iii is an *epistemological* claim – by what means could you tell that it is the same sensation type again.
    b. Well, what could I use if not my memory? But suppose my memory is faulty, then there will be, at the moment, no way to correct that. No one else knows what sensation I am naming. So, if I make a mistake, we may be stuck with that forever.
    c. But what the argument needs is a metaphysical claim: there cannot be a fact of the matter about whether this was S again or not.
    d. The transition from iii to iv says that if the epistemological point is true, then the metaphysical point is true.
    e. But that doesn't follow: it doesn't follow from the fact that some fact is not independently verifiable that there was *never a fact of the matter* about it to begin with.
        i. To suppose otherwise is to endorse a crude *verificationism* according to which there can only be facts if they can be verified.
        ii. No one should hold such a view: it's overwhelmingly plausible that Goldbach's conjecture is either true or false, but it may also be the case that it will never be proven or disproven.
12. But the only thing that matters for the question about LLMs to be factual is whether there is a fact of the matter; not whether it can be verified.
    a. And there will be such a fact of the matter if, in the initial baptism, I did zero in on a specific sensation type and named it.
    b. Nothing I have seen has shown me that that can't be done.
13. That's why I reject the PLA and why I can't go along with M's claim that our question is not a factual one.

Comments on Shanahan · Nate Sharadin · 5 November 2026 · bare page numbers are the manuscript's

- **1. Price's M-worlds:**
  - "the four Ms, for example: Morality, Modality, Meaning and the Mental": "important areas of human discourse seem threatened by the rise of modern science" (Price, "Naturalism and the Fate of the M-Worlds", 1997).
  - Price 2013, p. 5: "in many interesting cases it is hard to see what natural facts we could be talking about. Different people will offer different lists of these 'hard problems' – common candidates include meaning, value, mathematical truth, causation and physical modality, and various aspects of mentality".
  - Consciousness talk lives in one of them: the Mental.
  - **Morality:** the first M, and Price's own test case: "Some topic of human discourse–morality, say–seems difficult to accommodate in a naturalistic world view." (Price 1997, §1).
- **2. Why go pragmatist? Because the M-worlds seem suspicious!**
  - Eliminate them and we lose too much; defend them as non-natural and we need a faith in the limits of science. So "most philosophical naturalists look for some sort of rescue strategy" (Price 1997).
  - The pragmatist route "offers the benefits of nonnaturalism, without the metaphysical down side" (Price 1997); "no metaphysical spooks are thereby required." (Price 1997, §2)
  - **Morality:** the same menu, with names attached. Eliminate: Mackie, "There are no objective values", because "If there were objective values, then they would be entities or qualities or relations of a very strange sort, utterly different from anything else in the universe" (*Ethics: Inventing Right and Wrong*, 1977, pp. 15, 38). Defend as non-natural: Moore and the intuitionists.
- **3. So what does it mean to go pragmatist? A simple-minded pragmatist says (Blackburn):**
  - Pose the external question about a stretch of talk; answer it by saying what is done by talking that way, not by finding what it is about.
  - Blackburn, "Pragmatism: All or Some?" (2013), p. 75: "You will be a pragmatist about an area of discourse if you pose a Carnapian external question: how does it come about that we go in for this kind of discourse and thought? [...] Instead, the explanation proceeds by talking in different terms of what is done by so talking."
  - **Morality:** the view was built for ethics first. Blackburn, p. 75, straight after the definition: "All this is entirely in accord with the approach expressivists such as Gibbard and myself have taken to the ethical, which can encompass the more general area of the 'normative' ".
- **4. So: we should drop the demand to find something the discourse tracks; we should, instead, ask what it does.**
  - Shanahan says this himself! On p. 42: "metaphysically flavoured questions about what consciousness is have been supplanted by questions about how words like "consciousness", "awareness", "feeling", and so on are used."
  - **Morality:** Price 1997, §4, says the same of moral talk: "the remaining puzzle is merely sociolinguistic, and not metaphysical. It is the issue as to why creatures like us should find ourselves engaged in this particular linguistic practice, not the issue as to what these moral properties are."
- **5. What does the language do? Shanahan tells us:**
  - Prediction, p. 6: "To describe, predict, and explain complex behaviour, it often makes sense to adopt what Dennett calls the intentional stance".
  - p. 33 fn 88: to extend the language of consciousness to something is to "call upon the resources of those language games to **describe** the thing, to explain its past behaviour, to predict its future behaviour, and so on."
  - Fellowship and treatment, p. 33: the octopus "should be admitted to the fellowship of conscious creatures, and treated accordingly."
  - Attitude, not opinion, p. 33: "we are not "of the opinion" that the child or the horse is a sentient being. Things run much deeper than that. Unthinkingly, spontaneously, we simply take up the attitude towards them that we adopt towards fellow conscious creatures." (fn 89: PI II iv)
  - Moral standing, p. 32: "the capacity for consciousness goes hand-in-hand with candidature for moral consideration".
  - **Morality:** the expressivist's answer for moral talk has the same shape: its function is "to give voice to certain mental states, distinguished by their motivational role in human psychology" (Price 1997, §4). And the two practices aren't just parallel, they're joined: admitting something to the fellowship of conscious creatures is already a moral act (p. 32).
- **6. But then Shanahan looks for more than this:**
  - p. 10: "How is this space structured? What are its dimensions of variation, and where are its boundaries?"; the "compendium" of specimens, like "19th century collectors".
  - As if there were an answer to be had apart from: should we extend these practices to this thing?
  - He resists "Surely (the thought goes) there is a fact of the matter here." (p. 8), but only by going to "nothing IS anything" (p. 16) and "not ultimately real" (pp. 16–19, 25–27).
  - When he does reach "should we extend?", he puts it outside reason, p. 46: "the license to extend the language of consciousness to AI cannot be witheld [sic] on empirical or rational grounds"; "So the decision whether to extend the language of consciousness to AI is indeed a matter of existential orientation".
  - pp. 46–47: "we now set aside the concepts of truth, reality, rationality, etc, in favour of ambiguity, metaphor, imagery, allegory, story, polysemy, paradox, contradiction, etc. There is no law of the excluded middle when it comes to existential orientations."
- **7. Diagnosis: Shanahan is hostage to the idea that language is representational.**
  - So, since there's no hidden thing to represent, it must be that the represented thing is mysterious / empty.
  - For this reason, a question with no represented thing behind it must be beyond reason!
  - **Morality:** each of the moves in point 6 has a familiar moral twin:
    - The specimen collection: the moral realist's catalogue of the moral facts, as if there were a list to be had apart from what we should do.
    - "Surely ... there is a fact of the matter here" (p. 8), answered with "not ultimately real": Mackie's error

there are none really" (Blackburn's statement of the view he rejects, "Quasi-Realism No Fictionalism", 2005, §1).
    * The capitals: "nothing IS anything" (p. 16) is the "metaphysically heavyweight conception of what it would be for there to be moral truths REALLY" that Blackburn says isn't coherent (*Ruling Passions*, 1998, p. 319).
    * "Existential orientation": the existentialist's leap, which Blackburn calls "romantic, existential, fancy to imagine ourselves creating them by context-free acts of pure, normatively uncontaminated choice" (*Ruling Passions*, p. 257). The moral case shows the leap isn't what's left when the metaphysics goes.
- **8. What Shanahan needs is therapy.**
  - Patient: the urge to keep looking after the answer is already in.
  - And he cites the therapeutic Wittgenstein approvingly, as his own model (pp. 14, 56; p. 29 fns 79–81).
  - Why, for a pragmatist of his sort, it's a mistake to keep looking:
    * Saying what the talk does (point 5) isn't a first step towards the real story; it *is* the story. The account "eschews any use of the referring expressions of the discourse" (Blackburn): there's no second, deeper account of what consciousness talk is about waiting behind it.
    * So there's nothing left to look *for*. A space of minds with "dimensions of variation" and "boundaries" (p. 10) is exactly the subject matter the pragmatist said the talk doesn't track. Mapping it is metaphysics coming back in through the side door.
    * And there's nothing to *lose* by not finding it. Not finding it isn't a discovery that consciousness is "empty" or "not ultimately real" (pp. 16–19, 25–27); finding it was never what the talk was for. To be disappointed is to have kept the representationalist's expectations.
    * What's left are questions *inside* the practice: is the octopus in pain? should we admit this thing to the fellowship? These get answered the ordinary way, with reasons. That is what "leaves everything as it is" means: the fly is out of the bottle at point 5.
  - **Morality:** this is exactly the quasi-realist's point, and the moral case shows how it goes:
    * Asked whether he only defends our right to talk "as if" there were moral truths: "No, no, no. I do not say that we can talk as if kicking dogs were wrong, when 'really' it isn't wrong. I say that it is wrong (so it is true that it is wrong, so it is really true that it is wrong, so this is an example of a moral truth, so there are moral truths)." (Blackburn, *Ruling Passions*, p. 319)
  - Price 2013, p. 16 fn 5: "the subject naturalist recommends a dose of linguistic therapy: think carefully about what you are assuming about language before you allow yourself to be convinced that there's a genuine ontological puzzle."
  - Blackburn, "Truth, Realism, and the Regulation of Theory", *Essays in Quasi-Realism* (1993), p. 34: once the quasi-realist has earned truth, knowledge and the rest, "The realist never intended to say more than this, nor should the anti-realist have settled for less. [...] If this is so, philosophy indeed leaves everything as it is."
  - So I'm genuinely confused about why he wants to give us the things he is trying to give us: the map of the space of possible minds (p. 10), the emptiness ("nothing IS anything", p. 16), the existential orientation and the poetry (pp. 46–47). By his own Wittgenstein there is nothing left to explain, and the discovery is the one that lets us stop.
  - His own reason for going on is the mistake, p. 30: "the post-reflective condition ought to be liberating, not a shutting-down but an opening-out. Indeed, there are plenty of ways to respond to philosophy that treat it not as a source of questions demanding answers, but as a call to the poetic."
    * That treats the pragmatist's quiet as a shutting-down that something else has to open back up. But pragmatism doesn't shut down the conversation. It shuts down one question, the metaphysical one, and leaves every other question standing.
    * Is the octopus in pain? Should we admit it to the fellowship? Should we extend the language to an LLM, given what it's for, and what it costs to extend it wrongly or withhold it wrongly? All of that stays open, and it is argued over with reasons, not handed to poetry.
    * **Morality:** quasi-realism doesn't end moral argument either. Once the question of moral truths "REALLY" is dropped, we go on arguing about what's wrong, exactly as before; that's the point of earning "true" and "really" inside the practice.
    * So there was nothing to open out *from*. What is the opening-out *for*?
- **9. Why does he think he needs more? Two competing explanations.**
  - **(a) He thinks the pragmatist story debunks.** If consciousness talk is only for describing, predicting, admitting to the fellowship and treating accordingly, then nothing backs it, so it is "not ultimately real" (pp. 16–19, 25–27). He needs more to say where that leaves us, and what he finds is emptiness.
    * The tell: for animistic talk he wants a third option, "neither that trees and rocks speak in exactly the way that humans do, nor that they speak in a merely metaphorical sense (and hence, from a literal point of view, do not really speak at all)" (p. 44, quoting Burley).
    * The reply: that third option is quasi-realism. The "really" and the "true" are earned inside the practice; no metaphysics is needed to vindicate it, or to empty it.
    * **Morality:** this is the oldest objection to expressivism in ethics. Price 1997, §1: "Noncognitivism appeals most in the moral case, but even here some feel that it devalues the talk it claims to save."
  - **(b) He thinks the practice runs out at the edges.** The language games grew up around embodied creatures: for LLMs, "their lack of embodiment is especially challenging" (p. 42), and "the way LLMs are built and deployed imposes structural constraints on what someone can coherently say about them" (p. 47). Where the practice doesn't settle "should we extend?", he takes it that something beyond the practice must; finding nothing, he hands the question to existential orientation, where "the jurisdiction of science and reason is limited" (p. 46).
    * The reply: an unsettled "should we extend?" is a practical question, and practical questions take reasons: what the talk is for (point 5), and what it costs to extend it wrongly or withhold it wrongly. Not empirical, not logical, but not beyond reason either.

Understanding in Large Language Models

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The Question

(Question) Are LLMs genuinely intelligent, or are they just super-sophisticated, super-fast pattern-matchers that simulate intelligent behavior without being intelligent?

Do LLMs solve problems in ways that are similar to the way humans do so:

i. *Understand* the problem being posed.
ii. *Desire* to answer it and *intend* to answer it truthfully.
iii. *Reason* about what it would take to answer it correctly.
iv. *Know* whether the resources for answering it are already available to them or whether new evidence is required.
v. *Act* on those beliefs, desires and intentions to develop an answer.
vi. *Communicate* that answer by *telling* the interlocutor what the answer is.

(Q) Do LLMs solve problems the way humans do, or are they just super-sophisticated, super-fast pattern-matchers that simulate human behavior?

Our Question, Conceptual Engineering and Apt Methodology

Perhaps what you say is true about understanding, say, in its *full human version.* But the crucial question that needs answering is not Q but rather Q*:

(Q*) Do AIs instantiate the *bare essence* of these typical human capacities? It might be interesting to ask whether LLMs are able to instantiate the full human version of, say, understanding, but it's even more important to ask whether they are capable of *understanding per se*, even if it's not in its full human version – the bare essence of understanding, as we may put it.

(Q**) Do they instantiate concepts of understanding that are specially *engineered* to suit the new kind of intelligence they represent?

However, I would insist that the most responsible and fruitful way to approach these issues is to start with asking about the full human case, that is, to start with Q itself.

The Main Arguments for the Opposing Views

Our Question has quickly become very polarizing.

Believers are convinced that LLMs understand, know, believe, desire, intend and reason (Hinton, Cappelen, Sutskever, Lederman). Skeptics (Marcus, Bender, Chomsky, Pearl, Pinker, Penrose) who are just as convinced that LLMs are simply very sophisticated, very fast pattern-matchers.

In favor of the Believer's thesis that LLMs have human-like mental capacities there is first and foremost the fact that they are so *impressive* at what they can do. Best explanation is mentalistic.

What speaks in favor of the Skeptic's position? Some Skeptics argue as follows:

i. LLMs are at bottom stochastic next-token predictors.
ii. Such processes look nothing like belief, desire or intention.

ii. Therefore, LLMs don't have beliefs, desires or intentions.

This would not be a good argument. It doesn't work because (i) and (ii) don't entail (iii). *Emergence* is a real phenomenon:

A stronger consideration in the Skeptic's favor is that it's hard to vindicate the claim that lies at the *foundation* of the Believer's mentalistic explanation, namely:

> (Meaning and Understanding, M&U): LLMs *understand* what they are being asked and respond *meaningfully* to those questions.

Why would M&U be hard to vindicate? The *Grounding Problem*: LLMs don't have sense organs through which they interact with their environment (Harnad, 1990).

Meaning in LLMs

There is a view about intentionality which would not only easily get around the Grounding Problem but also swiftly decide the question in favor of the Believers: *Interpretivism*.

According to Interpretivism (see Dennett, 1987), a system has such intentional states as beliefs and desires provided it is easily *interpretable* as having those intentional states. And a system is easily so interpretable provided that thinking of it as having such beliefs and desires helps explain and predict its behavior in a certain range of cases.

The trouble is that Interpretivism is implausible. Many things are readily interpretable as having beliefs and desires, but which it would be absurd to say really have them.

A deeper problem is that 'interpretation' is itself an intentional state, so it's hard to see how something's being an intentional state could be *explained* via its being so interpretable.

Derivative vs Original Meaning.

Since the 1970s, philosophers have been interested in the ways in which our words can sometimes mean things not because we put that meaning into them, but because we stand in certain relations to those who did.

Understanding and Grasping Rules

Can we say that the LLMs *understand* what they are talking about? Like most philosophical words, the word 'understanding' has been used in different ways. In particular, it is natural to think that understanding comes in degrees.

In the *weakest* sense, you understand a word W when you can use W with its meaning M: it expresses M in your mouth.

In a *medium* sense, you understand W when you can not only use it to mean M but you also 'know what W means, know that it means M.' -- I've put this phrase in quotation marks because it's not clear what it itself means and I will come back to that.

In the *strongest* possible sense, you understand W in the way an expert does. For example, a high school student might understand addition in the medium sense, but not understand it as well as a number theorist might, who would know that it is a commutative, associative, cancellative function.

The sense in which I will be interested in LLM understanding is whether they understand in the medium sense, which we will need to characterize.

We're interested in the minimal notion that will let the Believer's story go through. But what notion of understanding is this? What is it for X to know what 'plus' means, or what it applies to?

Cappelen & Dever think that whatever it is, and we don't need a view about that just now, we can just *see* that LLMs understand in at least this sense. Can we just *see* this?

Even without a deep characterization of understanding, I think it's clear that understanding is not a perceptual matter in the way that C&D claim. In fact, given the way in which C&D defend the claim, I'm not sure *they* believe that it is a purely perceptual matter.

Let's distinguish sharply between two phenomena that can be easily conflated. The first is that some fact is established purely observationally. For example, I can know, on the basis of vision alone that Herman is wearing a (pink) shirt. On the other hand, the fact that the shirt is made in China I don't know purely perceptually, but I can often infer it from the fact that most shirts are made in China (let's say).

Now, sometimes these inferences are so obvious that they are made automatically. In such cases, it is easy to think that the knowledge is purely perceptual, even when it is inferential.

C&D give the following eg by way of supporting the perceptual thesis. Suppose you're trying to establish whether someone understands Mandarin. The best and easiest think you can do is get someone who understands Mandarin to speak to them for a bit. The Mandarin speaker would be able to establish quickly, without examining the interlocutor's innards, whether he understands Mandarin. So, whether someone understands Mandarin can be ascertained observationally.

I think this reasoning conflates the two phenomena distinguished above. If the speaker were really establishing the claim of understanding purely observationally, like in the case of color, they would need just *one* utterance from the interlocutor to establish whether understanding was present. But that's not what we do. We need to observe their speech for a bit.

That is not perception; that is *inference to the best explanation.* On the assumption that you are not aware of things that can converse fluently in a given language without understanding that language, you are very strongly entitled to conclude that the person understands Mandarin if they speak it fluently. However, if you become aware that the government has invested heavily in designing super-fast humanoid robots that can pass the Turing test in Mandarin, but which are equipped only with massive look up tables so as simulate conversing with understanding, you might well hesitate to make that inference.

Is the best explanation for what the conversational abilities of LLMS that we're seeing that they understand in the way humans so, or is it the alternative explanation via pattern matching?

I will try to make the case that the pattern matching explanation is the better explanation.

To evaluate the two hypotheses requires getting a little clearer about what they respectively say. What notion of medium understanding is at issue?

One of them we have already stated. To understand something is to have a capacity that's the *source* of one's competence with a word. What kind of thing could this source be?

You might think it could be: Being able to *state* correctly what the word applies to. Not that.

The most natural alternative is that it consists in some kind of ability, or disposition, to apply the word correctly. This is closer, but what does this in turn mean?

One thing it can't mean is that understanding in this sense consists in knowing how to apply the word correctly in *all possible* circumstances.

How should we think of it then? Should we perhaps say that it consists in a high degree of reliability in correct applications, setting the threshold at some appropriate number? 70%?

But that, too, would be absurd. Most of us learned the concept of addition at an early age. We are full masters of what it is to add two numbers together to get their sum. But can we be said to be reliable in 70% of all possible addition problems? Of course not.

As Kripke observed, the intuitive notion of understanding a word seems to demand construal as (something that either is or grounds) *grasp* of a *rule* for the correct application of the word. When you first come to understand, say, addition, this grasp of the rule for addition snaps into place.

What is crucial to recognize is that something like 'flash grasp' construal of understanding, in the full human case, is crucial to a number of important features that we take our speaking and thinking to have, features without which we would have to radically revise what we take ourselves to be doing.

First, words you understand have *infinitary* satisfaction conditions. For example, your word 'plus' applies in just this way and not that way to any two numbers, no matter how large. They are the satisfaction conditions for *your* word 'plus.' A word with a slightly different satisfaction conditions would be a different concept. Quus. There are an infinite number of such alternative functions.

How did your word 'plus' come to have these satisfaction conditions? The flash grasp picture explains it: when you understand 'plus' as meaning *plus*, you grasp some finitely stateable general rule, which you associate with the word, which determines *how* the word is to be *correctly* applied in an infinite number of cases. Nothing apart from a general rule could do that.

Second, when you competently use a word you understand, your use of the word is *explained* by your understanding.

Third, when you apply a word you understand, your application of the word is not only explained but *justified* by your understanding of its meaning. Suppose I mean *green* by 'green,' and, on seeing a green thing under good lighting conditions, I utter the sentence 'Here is a green thing.' When I say this, what I say is not only correct, given what I mean by 'green,' but is also *justified* by my grasp of the meaning of 'green.'

The important point is that in the full human case language use is a *rational* activity. When I use the word 'green' to describe this wall, I have a *reason* for using that word as opposed to another, a reason that is provided by my understanding of the word. Call this the Justificatory Requirement.

What's the significance of the Justificatory Requirement?

There is a way of looking at the orbits of the planets on which they are following elaborate and infinitary orbital rules. The rules could even be said explain why they move as they do, in the way in which on some views, the laws of nature explain why things behave as they do. But what the planets don't do is follow their paths because they have a *reason* or justification for doing so.

I assume we can all recognize that the way in which we follow the rules of our language is different from the way in which planets follow rules. And that it is an activity driven by reasons is the essential way in which it differs from mere mechanism. These reasons are supplied by our understanding and are consciously accessible to us: I know why I am disposed to apply the word 'green' to this green wall.

This shows a connection between this notion of understanding and certain notions of consciousness.

Of course, we already have such a connection in the case of terms like 'green,' whose meanings do not seem graspable without experiential acquaintance with their referent.

But what this Kripkenstein-inspired argument shows is that even in the case of logical or mathematical words, where no such acquaintance requirement is in place, there is still a potentially strong connection between understanding and consciousness.

Testing for Grasp
We see whether they are able to apply the rule correctly in *easy* and moderately difficult cases, while allowing for a much higher rate of errors in harder cases.

Why do we proceed in this way? The answer is obvious. Typical humans have limited working memories and limited powers of concentration. When those limitations are irrelevant – in the easy cases – we expect near perfection from someone who has grasped the rule.

Understanding in LLMs: The Argument from Easy Cases
One obvious problem for attributing strong understanding to LLMs is that they seem to be capable of going wrong precisely in very easy cases. Some of these examples are old, but that won't matter. One of them is from the most recent OpenAI model.

LLMs have been known:

> To assert that 9.11 is greater than 9.9 on the grounds that 11 is greater than 9.
>
> To be distracted by superficial linguistic cues. Example: "A farmer has 17 sheep. All but 9 die. How many remain?" Most people get that 'all but 9 die' means 9 remain. LLMs sometimes compute: $17 - 9 = 8$.

This is from GPT6.

> My prompt: Please multiply: \(5647 \times 8911
>
> \(5647 \times 8911 = \boxed{50,322,417}\)

This answer is incorrect.

I asked whether it was using a dedicated calculator or generating answers using the model itself? – No.

It added:

> I can use external computational tools. In this environment, I have access to Python, whi ch can perform exact integer arithmetic. Had I used Python to multiply 5,647 by 8,911, t hat would have involved an actual arithmetic computation executed outside the language model.
>
> Without such a tool, accuracy is not guaranteed. Language models can make arithmetic m istakes, especially with large numbers, because their learned computational procedures ar en't necessarily as systematic or reliable as conventional algorithms.

Interestingly, in the course of answering this question about method, it discovered that it had given the wrong answer.

> There's also a revealing detail: my previous answer was incorrect.
>
> The correct result is:
>
> \[5647 \times 8911 = \boxed{50,319,417} \]
>
> I had given 50,322,417.

Interestingly, this answer, too, is incorrect. It continued:

> This raises precisely the philosophical question you've been exploring: whether a neural n etwork that develops an internal procedure for multiplication should count as grasping th e rule of multiplication, even if its implementation is less reliable than that of a classical c omputer.

I said: actually, the correct answer is 50,320,417. ChatGPT6 responded:

> You're quite right.
>
> $$5647 \times 8911 = \boxed{50,320,417}$$
>
> My first answer was wrong, and so was my attempted correction. Your answer is correct. This is a particularly striking illustration of the difference between generating plausible arithmetic answers and reliably executing an arithmetic algorithm.

It's very plausible that if these models really had an understanding of what they're being asked, there would be no mistakes of this type.

Saying 9.11 is greater than 9.9 on the grounds that 11 is bigger than 9, if said by a child, would show that grasp of the concept of decimal notation had not yet snapped into place.

And the same goes for the multiplication case.

Reply to the Problem of Easy Cases: Humans Make Simple Errors, Too
There are two possible replies to this line of argument. The first one notes that even humans are capable of elementary errors consistent with their understanding the subject at hand, so we've set the bar too high.

Look at the multiplication case. If a human were to make an error in such a case, it would be due not to their not knowing what multiplication is, but to a performance error, due to memory and attention limitations. But these limitations don't apply to LLMs with their massive compute power and blazing speed.

Given a choice between two different types of explanation for these errors:

- understanding plus performance limitations versus no understanding, and
- being misled by patterns in the training set –

it does seem as though the latter is the more plausible explanation in the LLM case, while the former is the more plausible explanation in the human case.

Reply: These Errors Will Eventually Disappear
I am doubtful that there will come a time when even the *possibility* of such elementary errors will have been decisively eliminated. Easy errors arise because they are *outlier* cases. I assume that, so long as the training database is finite, there will always be outlier cases where the system can be tempted to behave in these uncomprehending ways. That's why I'm doubtful that the possibility of such errors can be definitively eliminated.

LLMs and Grasping Rules
But what if the possibility of all such errors *is* eliminated?

The main argument for not attributing strong understanding that doesn't depend on the possibility of easy errors is that there seems to be nothing corresponding to the 'snapping into place of the grasp of a rule' that then stands in extension-determining, explanatory and justificatory relations to a potential infinity of subsequent uses that we saw to be characteristic of the human case.

There are people who think that there is evidence that LLMs cotton onto rules as is shown in the phenomenon known as 'grokking.' Grokking is a phenomenon in machine learning where a neural network first appears to memorize its training data and later seems to suddenly discover a general rule that lets it perform well on new cases.

There are two questions here:

(1) Does the network come to represent something general, rather than merely memorize instances?

(2) And does that amount to grasping a rule in the sense in which a human grasps one?

On the first question, there is quite strong evidence that, at least in some cases, the answer is yes.

But can this really be said to be an infinitary rule with an infinite domain? Ordinary addition is a function on the natural numbers, so there are infinitely many possible cases.

Most classic grokking experiments use finite domains such as addition modulo 97. Perfect generalization there does not by itself demonstrate mastery of an infinite extension. A skeptic can say that the network has discovered an efficient compression of a finite function rather than grasped a rule governing infinitely many cases.

What could possibly determine about a particular LLM node that it instantiates the plus rule and not something that diverges from it at very large numbers? The usual ways of pinning down interpretability in non-humans – the designer's intentions, the telos of the thing (assuming it was a biological organism) – do not apply here.

Also in doubt would be the usual counterfactuals – if I were to extend the thing's lifetime, it would do such-and-so; unlike ordinary devices, like calculators, or symbolic computers, we are much less clear about these counterfactuals. The plus/quus problem is a real one for these networks.

Even putting this issue about finitude to one side, we would still be missing the Justificatory requirement. For there seems to be nothing in an LLM that amounts to acting on a reason, to the machine's having a justification for giving one answer as opposed to another.

Yet justification is what turns it from mere mechanical action to something that's rationally assessable. Rational assessabilty requires doing something for a reason, and not just mechanically. And that in turn seems to involve treating something as a reason and doing something because of that. (This structure is also present in reasoning itself.)

Is there a More Deflationary Understanding of Understanding That's Still Adequate?

The question is: What notion of understanding would be left if we gave up both on infinitary extensions and on language use being a rational activity? These are the questions that need to be thought through.